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## 1 Lab Title: Process Management

### 1.1 Introduction

Process Management is one of the main function of an operating system. It deals with creating, running, pausing and ending process of a program which are running in the computer. It also coordinates and manage the process to ensure efficient CPU utilization which also makes smooth system performance.

The main function of process management is to ensure the computer always runs smoothly and efficient. It supports multiprogramming and multitasking systems so the program runs only one process at a time. There are two types of process in process Management CPU Bond and IO/Bond. CPU Bond Process means that types of process where it spends most of the time in CPU for mash amount of calculation. For example playing a 3D game need a mash amount of calculation which takes place in CPU. IO/Bond

Process is that process which spend more time for waiting input and output operations. For example downloading a file which includes some input operations in the device. CPU process is faster than the IO process where CPU can calculate operations very faster whereas Input Output take some time for the operation.

The kernel which is the core of operating system which acts as a bridge between hardware and software manages these processes.

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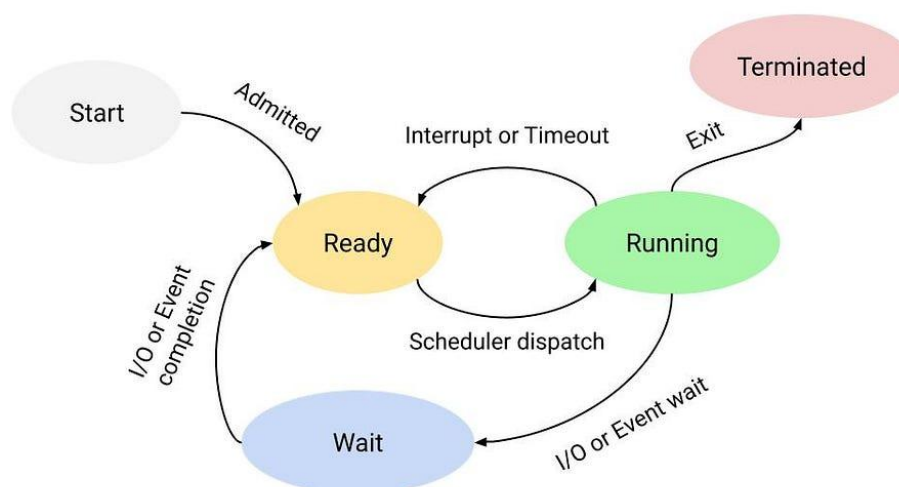


Figure 1: Process Management

## 2 Objective

The main objective of Process Management are as follows :-

- Kernel handles in process management as it creates, schedules and terminates processes.
- Kernel allocates and deallocates the memory as per the need.
- Kernel acts as a bridge between hardware and software allowing communicate to hardware devices.
- Kernel helps in managing the file system by providing a file system interface to applications.
- The kernel allows processes to communication with each other using mechanism like pipes, signal etc.

### 3 Types of Kernel

A kernel is a computer program which is the brain of operating system that has complete over everything in the system. It acts as the bridge between hardware and software. It controls computer hardware components like I/O, memory and also optimize the use of common resources such as CPU, cache, file system and network sockets. A kernel is one of the first program loaded in the computer after the bootloader. There are many types of kernel used in modern operating system.

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#### 3.1 Monolithic kernel

A monolithic kernel is an operating system where operating system run in a kernel space that means they all share the same memory space. A kernel space means where the operating system and other important parts run.

Advantage of Monolithic kernel in Operating System

- It provides CPU scheduling, memory management, file management functions through system calls.
- It is a single static binary file.
- It is a single large process running in a single address space which boosts or increase the fast.

Disadvantages of Monolithic kernel in Operating System

- If any system of Monolithic kernel service is full then it leads to entire system failure.
- If the admin wants to add any new service then the entire operating system need to be modify.
- A bug or virus is seen in any system would effect the whole system.

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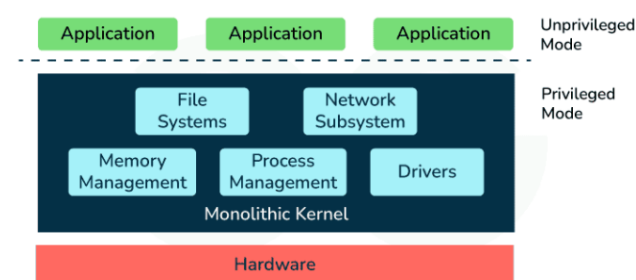


Figure 2: Monolithic kernel

### 3.2 Microkernel in Operating System

A microkernel is a type of operating system kernel where some basic services run in kernel space where other service running in user space. User space is where user application run.

#### Advantages of Microkernel in Operating System

- The kernel and services developed and maintained independently so it can build, fixed and updated easily without operating the whole system.
- The kernel contains some essential functions required to manage the system improve performance and also make system faster and more efficient.
- The kernel can also improve security complex.
- The microkernel are less complex than the Monolithic kernel.
- The microkernel can be moved to different platforms with less effort.

#### Disadvantages of Microkernel in Operating System

- The kernel message passing between user-level processes can effect performance, mainly in high-performance application.
- The modular design increases the complexity which makes more difficult to develop and maintain the Operating System.
- Separation of kernel and user-level, results limited performance optimization.
- Consume more memory compared to a Monolithic kernel.



Figure 3: Microkernel

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### 3.3 Hybrid kernel

A hybrid kernel is a kernel combining both aspects of microkernel and monolithic kernel used in computer operating system. From monolithic kernel it takes the performance advantage and modularity and stability from the microkernel. The main service like memory management , process management , system calls run in hybrid kernel but other services like device driver, file system may run in user space which balance the system as it is neither too heavy like monolithic kernel nor to slow like a micro kernel

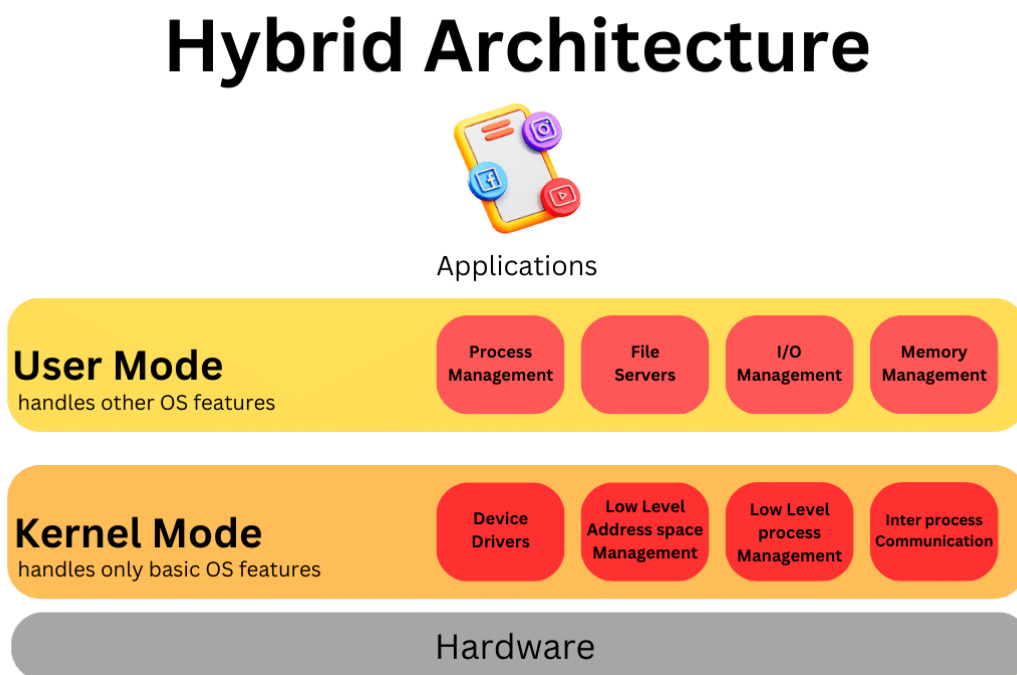


Figure 4: Hybrid Kernel



## 4 Conclusion

In Conclusion, the process management is the brain of an operating system. Without process management computer could not able to run multiple programs at the same time efficiently. CPU time is used efficiently and multiple programs can be used at a same time. Understanding CPU bond and I/O bound processing helps in scheduling task to improve the system performance.

The kernel plays vital role in process management as it is responsible in memory management, resource management, device management, error handling, task scheduling and load balancing. By studying these concept we can better modern operating system, including network operating system handle multiple calculations in a reliable and efficient manner.

## 5 References

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